Final Exam

● Graded

Student

Akbar Akramov

Total Points

44.5 / 45 pts

Question 1

Overall Grades **44.5** / 45 pts

✔ **– 0 pts** See the details on each problem.

💬 **– 0.5 pts** Point adjustment

MATH 202-001 FALL 2026 SESSION 1 FINAL EXAM
OCTOBER 19TH, 15:30–18:30

Name (Please Print)	NetID	Signature
Akbar Akramov	aa921	

INSTRUCTION

- This is a 180-minute test.
- The total number of points is 45.
- **No Cellphones.**
- **No Calculators.**
- **Do NOT write on the back of the test; if you do so, your answer(s) will NOT be graded.**
- There are 9 problems.
- Show your work. To receive full credit, your answers must be neatly written, logically organized, and clearly labeled. Writing just the answer will not give you credit for that.
- The backs and extra pages are offered as scratch paper.
- GOOD LUCK.

Duke Community Standard

Duke Kunshan University (DKU) is a community comprised of individuals from diverse cultures and backgrounds. We are dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. Members of this community commit to reflecting upon and upholding these principles in all academic and non-academic endeavors and to protecting and promoting a culture of integrity and trust.

To uphold the DKU Community Standard

- I will hold myself to the highest standards for honesty, integrity, fairness, and responsibility in my academic and non-academic endeavors
- I will respect other cultures and embrace all forms of diversity
- I will uphold the standards if they are compromised.

Honor Pledge: I will neither give nor receive aid on this test. SIGNATURE:

IF YOU DO NOT SIGN THE HONOR PLEDGE, YOUR FINAL EXAM WILL BE RECORDED AS 0 POINTS.

Problems	1	2	3	4	5	6	7	8	9	Total
Scores	5	4.5	5	5	5	5	5	5	5	44.5

Problem 1. [5] Find the general solution of the linear system:

$$\begin{cases} 2x - 3y = 1 \\ -2y - 3z = 1 \\ x + 2z = 1 \end{cases}$$

$$\left[\begin{array}{ccc|c} 2 & -3 & 0 & 1 \\ 0 & -2 & -3 & 1 \\ 1 & 0 & 2 & 1 \end{array}\right] \quad R_1 \leftrightarrow R_3 \quad \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & -2 & -3 & 1 \\ 2 & -3 & 0 & 1 \end{array}\right] \quad R_3 \leftarrow R_3 - 2R_1 \quad \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & -2 & -3 & 1 \\ 0 & -3 & -4 & -1 \end{array}\right]$$

$$R_2 \leftarrow R_2 - R_3 \quad \left[\begin{array}{ccc|c} 1 & 0 & 2 & -1 \\ 0 & 1 & 1 & 2 \\ 0 & -3 & -4 & -1 \end{array}\right] \quad R_3 \leftarrow R_3 + 3R_2 \quad \left[\begin{array}{ccc|c} 1 & 0 & 2 & -1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & -1 & 5 \end{array}\right] \quad \begin{array}{l} -z = 5 \\ \boxed{z = -5} \end{array}$$

$x + (-10) = 1 \Rightarrow \boxed{x = 11}$ $\qquad 2z - 3y = 1$ $\qquad \boxed{y = 7}$

4.5

Problem 2. [5] Let $A = \begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$. Find

(1) a basis of the column space of A,
(2) a basis of the row space of A,
(3) and a basis of the null space of A.

$R_4 \leftrightarrow R_1$, $R_3 \leftrightarrow R_2$ $\begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 1 \\ -5 & 0 & 5 \\ 2 & 4 & 2 \end{bmatrix}$ $R_3 \leftarrow R_3 + 5R_1$, $R_4 \leftarrow R_4 - 2R_1$ $\begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 1 \\ 0 & 15 & 15 \\ 0 & -7 & -2 \end{bmatrix}$ $R_3 \leftarrow \frac{1}{15} R_3$, $R_1 \leftarrow -\frac{1}{2} R_2$ $\Rightarrow \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

$\Rightarrow$ $R_4 \leftarrow R_4 - R_3$, $R_3 \leftarrow R_3 - R_2$ $\begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $R_1 \leftarrow R_1 - 3R_2$ $\Rightarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

1) basis of column space of A $\rightarrow \left\{ \begin{bmatrix} 2 \\ -5 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ 1 \\ 3 \end{bmatrix} \right\}$ $\rightarrow$ original pivot columns of A ~~in order~~

2) basis of row space of A $\rightarrow \left\{ \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$ $\rightarrow$ non-zero rows of reduced form

3) basis of the null space of A $\rightarrow \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$

Problem 3. [5'] Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ be a linear transformation such that

$$T\left(\begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad T\left(\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 2 \end{bmatrix}, \qquad T\left(\begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 2 \end{bmatrix}.$$

Find the matrix transformation of T, i.e., find a matrix A such that

$$T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = A \begin{bmatrix} x \\ y \\ z \end{bmatrix}.$$

$T(v) = Av$ by linearity $T(e_3) = t(2,2) = (-2,-2)$

$T(e_1) = T(1,0,-1) + T(e_3) = (0,2) + (-2,-2) = (-2,0)$

$T(e_2) = T(0,1,-1) + T(e_3) = (1,1) + (-2,-2) = (-1,-1)$

So $\Rightarrow$ $A = \begin{bmatrix} T(e_1) & T(e_2) & T(e_3) \end{bmatrix}$ $\Rightarrow$ $A = \begin{bmatrix} -2 & -1 & -2 \\ 0 & -1 & -2 \end{bmatrix}$

Problem 4. [5] Find the Jordan canonical form of

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$$

$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ → 3rd column nonzero. $\det(A - \lambda I) = (1-\lambda)\begin{vmatrix} 1-\lambda & 2 \\ 0 & 2-\lambda \end{vmatrix} =$

$\Rightarrow (1-\lambda)^2(2-\lambda)$ $\lambda_1 = 1$ → multiplicity 2 $\lambda_2 = 2$

Finding eigenvector for $\lambda = 1$

$A - I = \begin{bmatrix} 0 & 2 & 0 \\ 0 & 1 & 0 \\ -2 & -2 & 0 \end{bmatrix}$ $(A-I)x = 0$ $y = 0$ $x = 0$ z = free

$\dim(A - I) = 1$ $\lambda = 1$ has multiplicity 2 so we need Jordan block of size 2.

Since $\lambda = 2$ has algebraic multiplicity 1, it gives $J_2 = [2]$ 1×1

So [redacted] $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

Problem 5. [5'] Diagonalize the quadratic form

$$f(x,y,z) = 3x^2 + 3y^2 + 4z^2 + 2xz - 2yz.$$

$A - \lambda I = \begin{bmatrix} 3-\lambda & 0 & 1 \\ 0 & 3-\lambda & -1 \\ 1 & -1 & 4-\lambda \end{bmatrix}$ $\det(A-\lambda I) = (3-\lambda)[(3-\lambda)(4-\lambda)-1] - (3-\lambda)$

$= (3-\lambda)(\lambda^2 - 7\lambda + 11) - (3-\lambda) =$

$= (3-\lambda)(\lambda^2 - 7\lambda + 11 - 1) = (3-\lambda)(\lambda-5)(\lambda-2)$

$\lambda = 3, \ \lambda = 5, \ \lambda = 2$

$A - 3I = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & -1 \\ 1 & -1 & 1 \end{bmatrix} R_1 \leftrightarrow R_3 \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} R_3 \leftarrow R_3 - R_2 \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

$v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

$A - 5I = \begin{bmatrix} -2 & 0 & 1 \\ 0 & -2 & -1 \\ 1 & -1 & -3 \end{bmatrix} R_1 \leftrightarrow R_3 \begin{bmatrix} 1 & -1 & -3 \\ 0 & -2 & -1 \\ -2 & 0 & 1 \end{bmatrix} R_3 \leftarrow R_3 + 2R_1 \begin{bmatrix} 1 & -1 & -3 \\ 0 & -2 & -1 \\ 0 & -2 & -5 \end{bmatrix} R_3 \leftarrow R_3 + R_2$

$\Rightarrow \begin{bmatrix} 1 & -1 & -3 \\ 0 & -2 & -1 \\ 0 & 0 & 0 \end{bmatrix} R_2 \leftarrow -\frac{1}{2} R_2 \begin{bmatrix} 1 & -1 & -3 \\ 0 & 1 & 1/2 \\ 0 & 0 & 0 \end{bmatrix} R_1 \leftarrow R_1 + R_2 \begin{bmatrix} 1 & 0 & -5/2 \\ 0 & 1 & 1/2 \\ 0 & 0 & 0 \end{bmatrix}$

$v_2 = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$

$A - 2I = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 2 \end{bmatrix} \begin{matrix} R_3 \leftarrow R_3 - R_1 \\ R_3 \leftarrow R_3 + R_2 \end{matrix} \Rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$

$v_3 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$

$P = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{3} & -1/\sqrt{6} \\ 1/\sqrt{2} & -1/\sqrt{3} & 2/\sqrt{6} \\ 0 & 1/\sqrt{3} & 1/\sqrt{6} \end{bmatrix} \quad D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

$f(x,y,z) = 3t^2 + 5u^2 + 2v^2$

Problem 6. [5] Find a, b, and c such that $\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix}.$$

$$a\begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix} + b\begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix} + c\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix} \qquad \begin{cases} 2b - c = -3 \\ 2a - b + c = 8 \\ -2a + 2b = -8 \end{cases} \qquad \begin{array}{l} c = 2b + 3 \\ a = b + 4 \end{array}$$

$$\Rightarrow 2(b+4) - b + (2b+3) = 8 \Rightarrow 3b + 11 = 8 \Rightarrow \boxed{b = -1} \qquad \begin{array}{l} a = -1 + 4 = \boxed{3} \\ c = -2 + 3 = \boxed{1} \end{array}$$

$$\boxed{a = 3,\ b = -1,\ c = 1}$$

5/5

Problem 7. [5] Let V be a subspace of $\mathbb{R}^4$, defined by

$$V = span\left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}\right\}.$$

Find an orthonormal basis for V.

$v_1 = \begin{bmatrix}0\\1\\1\\0\end{bmatrix}$ $\quad \|v_1\| = \sqrt{0^2+1^2+1^2+0^2} = \sqrt{2}$ $\quad e_1 = \frac{1}{\sqrt{2}}\begin{bmatrix}0\\1\\1\\0\end{bmatrix}$

$w_2 = v_2 - \frac{v_2 \cdot v_1}{v_1 \cdot v_1} v_1$ $\quad v_2 \cdot v_1 = 0+0+2+0 = 2$ $\quad v_1 \cdot v_1 = 2$ $\quad$ thus $w_2 = \begin{bmatrix}3\\0\\2\\-1\end{bmatrix} - \frac{2}{2}\begin{bmatrix}0\\1\\1\\0\end{bmatrix} = \begin{bmatrix}3\\-1\\1\\-1\end{bmatrix}$

normalizing w_2 $\quad \|w_2\| = \sqrt{9+1+1+1} = \sqrt{12} = 2\sqrt{3}$ $\quad$ so $e_2 = \frac{1}{2\sqrt{3}}\begin{bmatrix}3\\-1\\1\\-1\end{bmatrix}$

The orthonormal basis for V is $\left\{\frac{1}{\sqrt{2}}\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \frac{1}{2\sqrt{3}}\begin{bmatrix}3\\-1\\1\\-1\end{bmatrix}\right\}$

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Problem 8. [5'] Let A be an $n \times n$ square matrix satisfying $A^T = -A$. Show that if n is odd, then $\det A = 0$.

$\det(A^T) = \det(A)$ since $A^T = -A$ $\det(A) = \det(-A)$

$\det(-A) = (-1)^n \det(A)$ since n is odd $(-1)^n = -1$ so $\det(A) = -\det(A)$

$\Rightarrow 2\det(A) = 0 \Rightarrow \det(A) = 0$

Problem 9. [5'] For two vectors $\vec{u}$ and $\vec{v}$, suppose they satisfy that

$$\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|.$$

Show that $\vec{u}$ and $\vec{v}$ are perpendicular to each other.

$\|\vec{u}+\vec{v}\|^2 = \|\vec{u}-\vec{v}\|^2 \Rightarrow \|\vec{u}\|^2 + 2\vec{u}\cdot\vec{v} + \|\vec{v}\|^2 = \|\vec{u}\|^2 - 2\vec{u}\cdot\vec{v} + \|\vec{v}\|^2$

$\Rightarrow 2\vec{u}\cdot\vec{v} = -2\vec{u}\cdot\vec{v} \Rightarrow 4\vec{u}\cdot\vec{v} = 0 \Rightarrow \vec{u}\cdot\vec{v} = 0$ the dot product is zero, so they are perpendicular.